# LACHLAN PASSENGER

lachlan.passenger1@monash.edu | lachlanpassenger.github.io

Monash University – Clayton, Victoria 3800 – Australia

## SUMMARY

I'm an astrophysicist who uses statistical methods to answer questions about the universe. I develop tools for studying gravitational waves and the merging black holes and other compact binaries that produce them. I'm a member of the OzGrav ARC Center of Excellence for Gravitational Wave Discovery and the LIGO–Virgo–KAGRA collaboration, where I contribute my skills to the global effort to understand these systems. I'm a PhD candidate at Monash University, where I'm advised by Eric Thrane and Paul Lasky.

## EDUCATION

- **Monash University** — *2024 – Present (submitting Aug 2027)*
  *Doctor of Philosophy (PhD), Astrophysics* — Melbourne, Australia
  - Thesis title (preliminary): Model testing in gravitational-wave astronomy
  - Supervisors: Eric Thrane, Paul Lasky

- **Monash University** — *Feb 2023 – Nov 2023*
  *Bachelor of Science (Honours), Astrophysics* — Melbourne, Australia
  - First class honours, GPA: 4.0/4.0, WAM: 87.0

- **The University of Queensland** — *2020 – 2022*
  *Bachelor of Science, Physics extended major* — Brisbane, Australia

## FIRST AUTHOR PUBLICATIONS

**[1]** **Lachlan Passenger**, Nir Guttman, Paul D. Lasky, Eric Thrane (2026). **How general relativity stacks up: Testing gravity by coherently combining gravitational-wave observations**. *arXiv:2610.29247*.

**[2]** **Lachlan Passenger**, Sharan Banagiri, Eric Thrane, Paul D. Lasky, Angela Borchers, Maya Fishbach, Claire S. Ye (2026). **Is GW231123 a hierarchical merger?** *The Astrophysical Journal*, Vol. 999, Art. 236.

**[3]** **Lachlan Passenger**, Shun Yin Cheung, Nir Guttman, Nikhil Kannachel, Paul D. Lasky, Eric Thrane (2025). **A Gaussian process framework for testing general relativity with gravitational waves**. *The Astrophysical Journal*, Vol. 994, Art. 202.

**[4]** **Lachlan Passenger**, Eric Thrane, Paul D. Lasky, Ethan Payne, Simon Stevenson, Ben Farr (2024). **Are all models wrong? Falsifying binary formation models in gravitational-wave astronomy**. *Monthly Notices of the Royal Astronomical Society*, Vol. 535, Issue 3, pp. 2837–2843.

## CO-AUTHOR PUBLICATIONS

**[1]** Mia Babatsikos, Fan Liu, Amanda Karakas, Ilya Mandel, **Lachlan Passenger**, Megan Bedell, Sharon Wang, Zimo Cheng (2026). **The Sun's chemical peculiarity: disentangling Galactic chemical evolution and planetary engulfment in solar twins**. Manuscript accepted for publication in *The Astrophysical Journal*. arXiv:2607.01699

**[2]** Shun Yin Cheung, **Lachlan Passenger**, Paul D. Lasky, Eric Thrane (2026). **A search for extra polarizations using a Gaussian process in Gravitational-wave Transient Catalog 3**. *The Astrophysical Journal*, Vol. 1001, Art. 130.

**[3]** Anuradha Gupta, . . . , **Lachlan Passenger**, . . . , B. S. Sathyaprakash (2025). **Possible causes of false general relativity violations in gravitational wave observations**. *SciPost Physics Community Reports*, Art. 5.

## Collaboration publications

As a member of the LIGO-Virgo-KAGRA collaboration, I am a co-author on all collaboration publications. Here I only list papers for which I have made significant contributions.

**[1]** The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration (2026). **GWTC-5.0: Population properties of merging compact binaries**. arXiv:2605.27226

## Conference and seminar presentations

– **Testing general relativity**. Gravity and Extreme Matter workshop, The University of Melbourne. (April 2026).

– **A Gaussian process framework for testing general relativity with gravitational waves**. GR24/Amaldi16, Glasgow. (July 2025).

## Teaching experience

- **Monash University** *2024 – Present*
  *Teaching Associate* Melbourne, Australia
  - ASP1010 – Earth to cosmos: Introductory astronomy
  - ASP1022 – Life in the universe: Astrobiology
  - ASP3012 – Stars and Galaxies
  - SCI1005 – Scientific research and communication

## Service

- **Referee:** *Physical Review D*
- **Collaboration review:** Results reviewer for *GWTC-5.0: Tests of General Relativity*

## Skills

- **General:** Python, linear algebra, Bayesian/frequentist statistics, sampling methods and optimisation, Gaussian processes, high-performance computing, signal processing
- **Gravitational-wave specific:** BILBY, GWPOPULATION